

Chemistry 151 Lab

Fall 2026

CHEM HALL 011

Instructors and TAs

- Instructor:
  - Dr. Aimee Morris
  - ammorris@fortlewis.edu
  - Student Hours:
    - Office: Chem Hall 146
    - M 9:00 – 10:00am
    - W 9:00 – 10:00am
    - Th 11:15am – 1:15pm
    - Or by appointment
- TA:
  - Greysen Horan
  - Senior, Health Science Major

Agenda for today:

- Information about lab
- Safety
- Pre-lab talk for Calorimetry
- Calorimetry Lab

Make sure you turn in the following before leaving lab today:

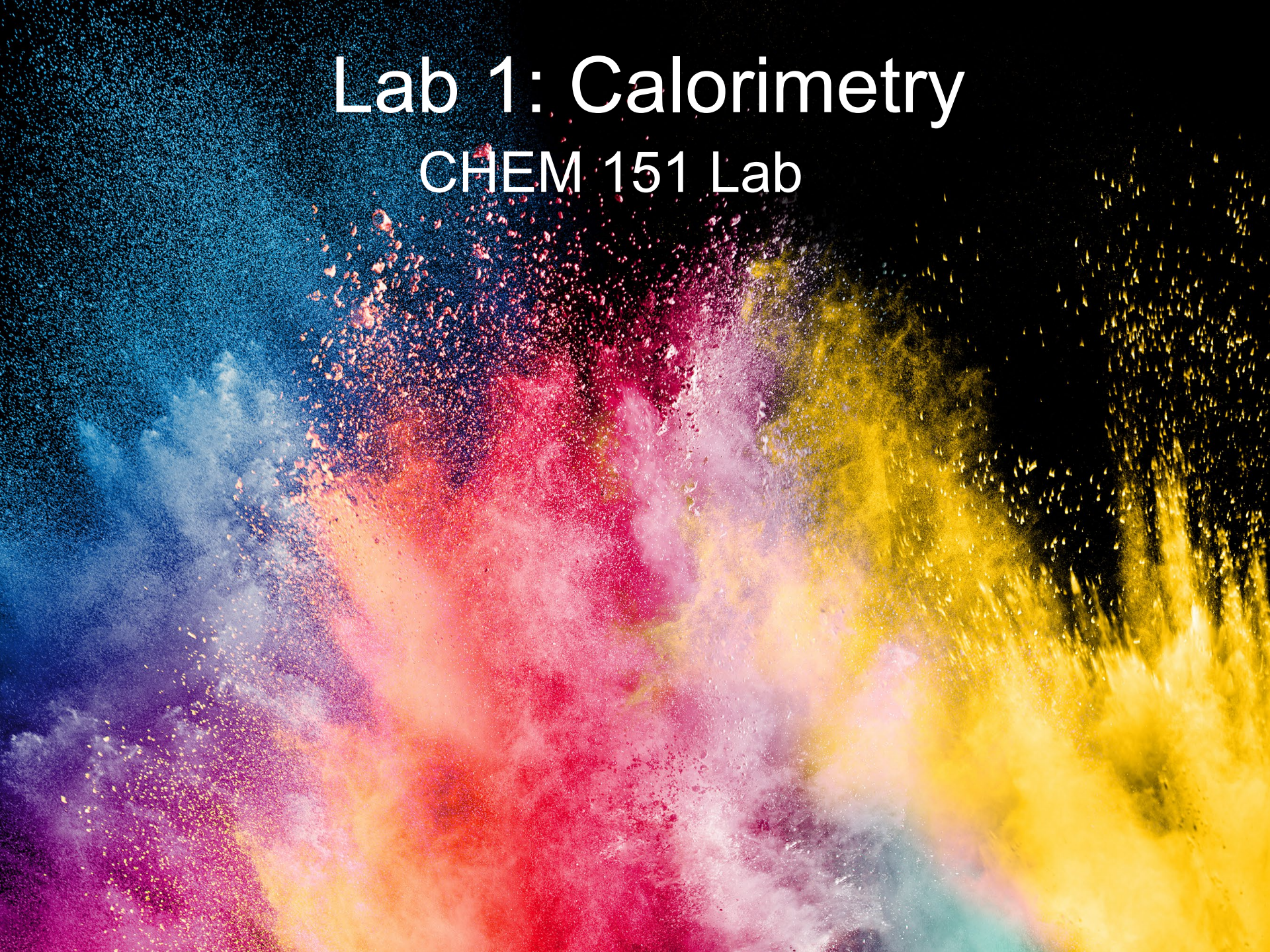
1. Signed Safety Agreement
2. Know how to electronically upload your Calorimetry/Thermochemistry lab notebook pages as a single pdf due 11:59pm on Wednesday

Safety

- Our number 1 priority in the lab is to ensure your safety!
  - Attire for lab
  - No food or beverages allowed
  - Proper handling of chemicals
  - Spills, fire, accidents
  - Cleanliness
- Read and sign the safety agreement and turn it end by the end of the lab period (5 points)
- Review the safety rules from Canvas page
- Come prepared to lab each week

Lab 1: Calorimetry

CHEM 151 Lab



Objectives

- Determine the specific heat of a metal to determine its identity

- Use the specific heat of water to determine the caloric content of food

Calculating the change in energy of a reaction

$$\Delta E = E_{final} - E_{initial}$$

$$\Delta E = E_{products} - E_{reactants} = q + w$$

Work (w) = $-P\Delta V$

- P = pressure (units of atm)
- $\Delta V = V_{final} - V_{initial}$

Heat (q) = $m \times C_s \times \Delta T$

- m = mass of the substance (grams)
- C_s = specific heat capacity of the object $\left(\frac{J}{g\ ^\circ C}\right)$
- $\Delta T = T_{final} - T_{initial}$ (°C)

The Units of Energy

1 calorie = heat required to raise temperature of 1.00 g of H_2O by 1.0 ^oC.

1000 cal = 1 kilocalorie = 1 kcal

1 kcal = 1 Calorie (a food "calorie")

But we use the unit called the JOULE.

1 cal = exactly 4.184 joules

Unit	Conversion
1 calorie (cal)	4.184 joules (J)
1 Calorie (Cal)	1000 cal or 1 kcal

If heat (q) = m x C_s x ΔT, what is C_s?

Heat Capacity:

The amount of energy (heat as q) required to raise an object's temperature by 1 ˚C.

$$q = C \times \Delta T$$

Heat capacity

Specific Heat Capacity (C_s):

The amount of energy (heat as q) required to raise ONE gram of matter's temperature by 1 °C.

$$q = m \times C_s \times \Delta T$$

Specific Heat Capacities(C_s):

A large C_s means that more heat is required to raise the temperature of that substance (e.g., water) by 1˚C than the heat required to raise an equivalent mass of a different substance (e.g., lead) that has a smaller C_s.

$$q = m \times C_s \times \Delta T$$

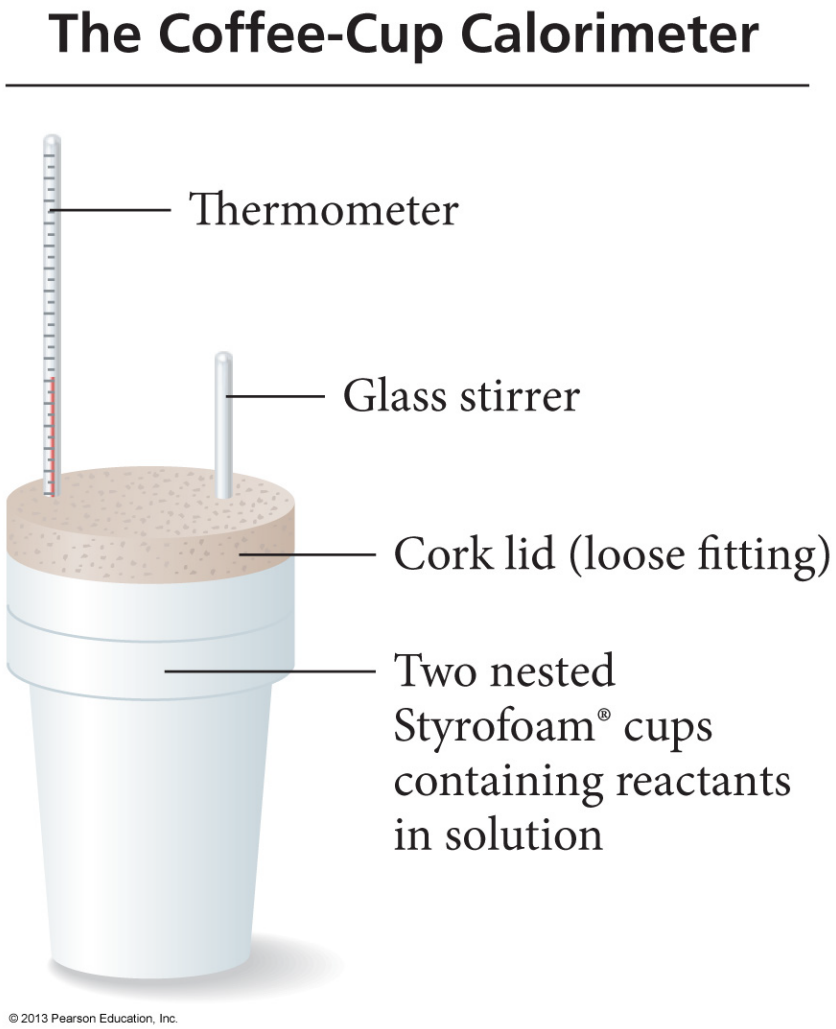
TABLE 6.4 Specific Heat Capacities of Some Common Substances

Substance	Specific Heat Capacity, C_s (J/g · °C)*
Elements	
Lead	0.128
Gold	0.128
Silver	0.235
Copper	0.385
Iron	0.449
Aluminum	0.903
Compounds	
Ethanol	2.42
Water	4.18
Materials	
Glass (Pyrex)	0.75
Granite	0.79
Sand	0.84

*At 298 K.

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Constant pressure calorimetry (open to the atmosphere)



A known mass of solution is used

During the reaction, the heat evolved (or absorbed) causes a ΔT in the solution

If the specific heat capacity of the solution is known, we can calculate q_{soln}

$$q_{soln} = m_{soln} \times C_s \times \Delta T$$

$$q_{soln} = - q_{rxn}$$

Procedure notes

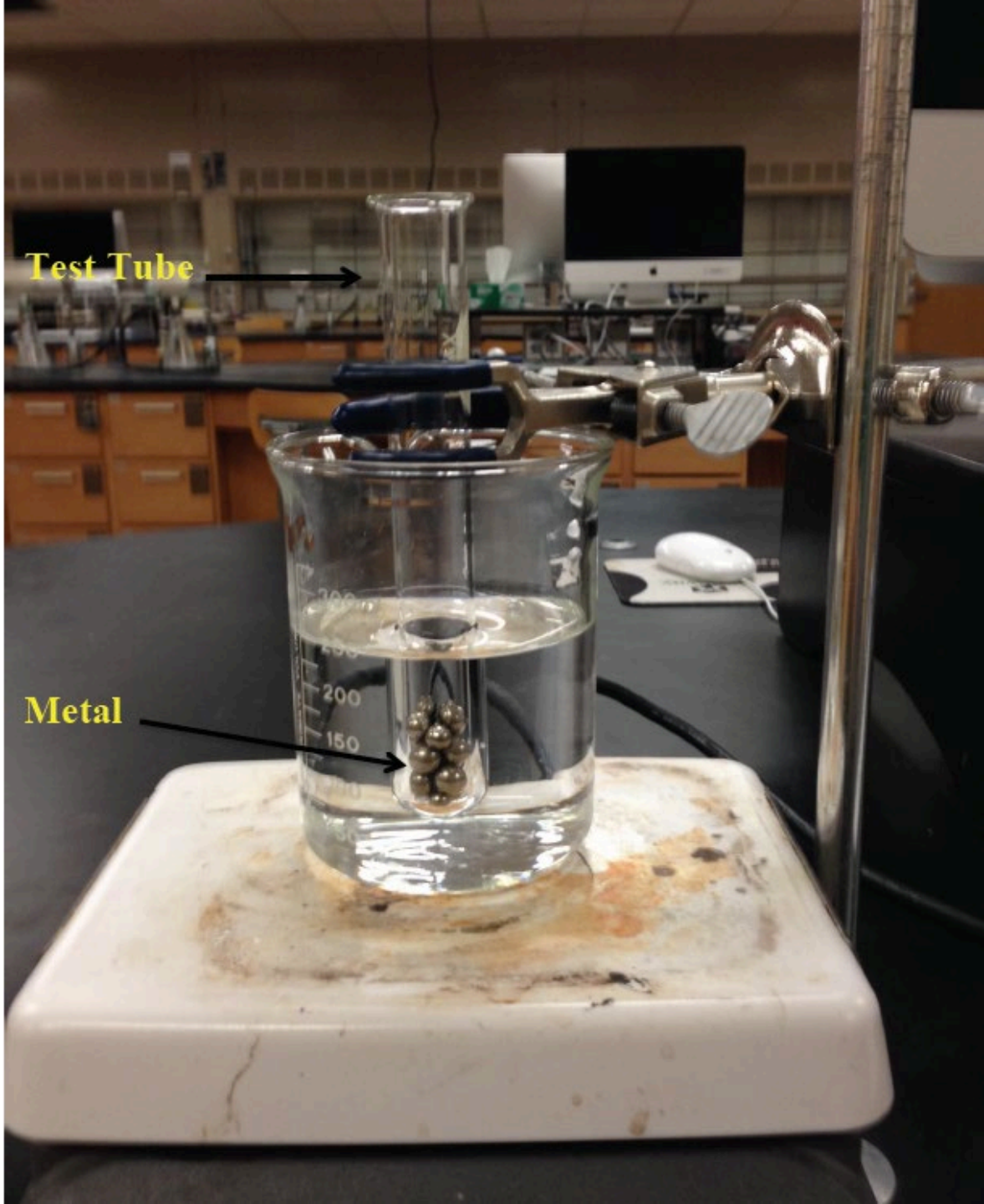
- **Part A:** Measuring the specific heat of two unknown metals to determine their identity
- **Part B:** Food is fuel

In this lab you will use the specific heat of water and its change in temperature to determine the caloric content in a Frito™ and Cheeto™.



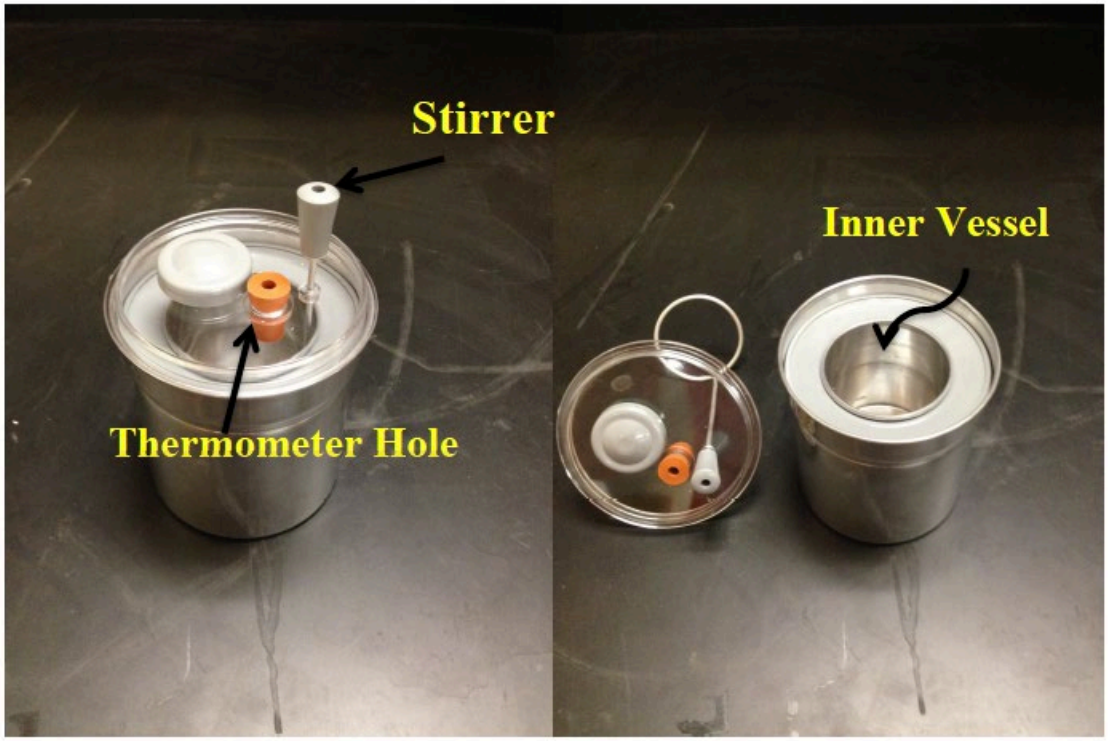
Part A setup

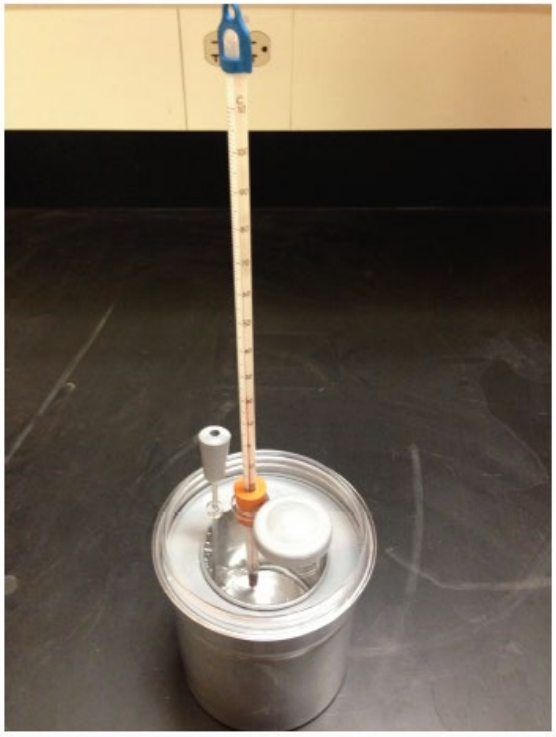
1. We will heat up the metal to the temperature of boiling water (94.5 ℃). This will be our $\mathbf{T_{initial}}$ **for the metal.**



Part A setup

2. Setup your constant pressure calorimeter.
    - Transfer 100.0 mL of water to the inner vessel.
    - Insert a thermometer into the calorimeter.
    - Record the initial temperature of the water. This will be the **$T_{initial}$ for the solution of water.**

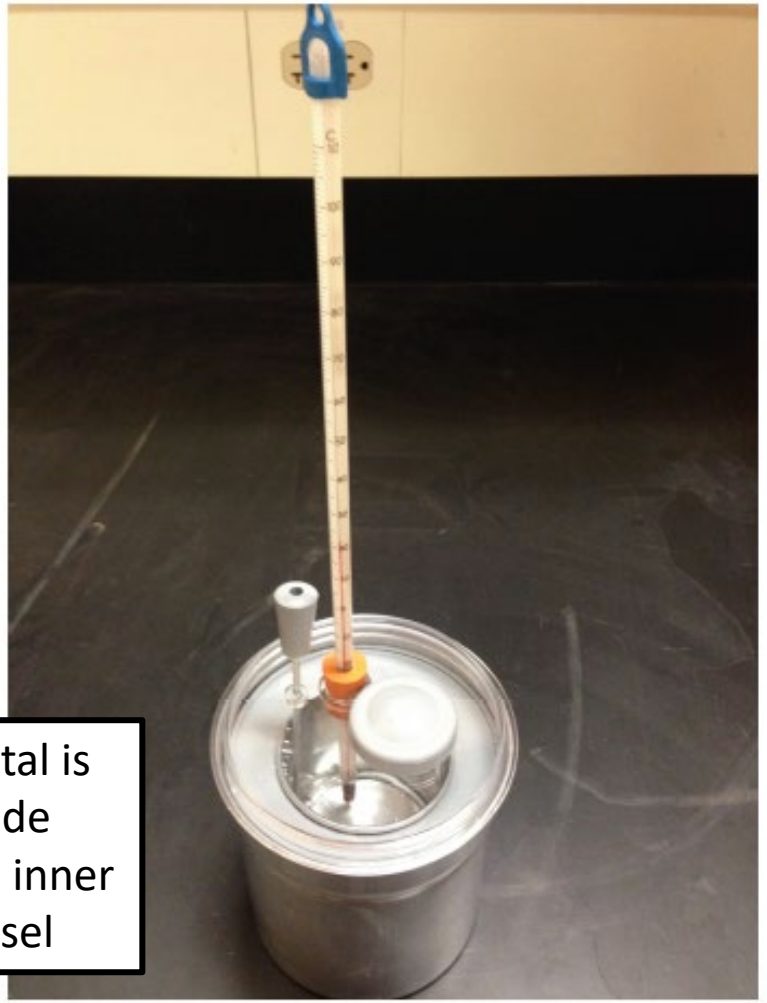




Part A setup

3. Quickly transfer your hot metal (from step 1) to the inner vessel of the calorimeter.

- Put the lid back on the calorimeter and gently stir.
- Watch the temperature slowly rise and record the maximum temperature reached. This will be the **T_{final} for the solution of water and for the metal.**



Metal is inside the inner vessel

Part A calculations

4. You know have everything you need to calculate the specific heat capacity of the metal ($C_{s(metal)}$):

$$q_{water} = m_{water} \; x \; C_{s(water)} \; x \; \Delta T_{(water)}$$

$$q_{water} = -q_{metal} = mmetal \; x \; C_{s(metal)} \; x \; \Delta T_{(metal)}$$

5. Use C_s to determine the identity of the metal using the Table in the lab manual

Part B setup – Food is fuel

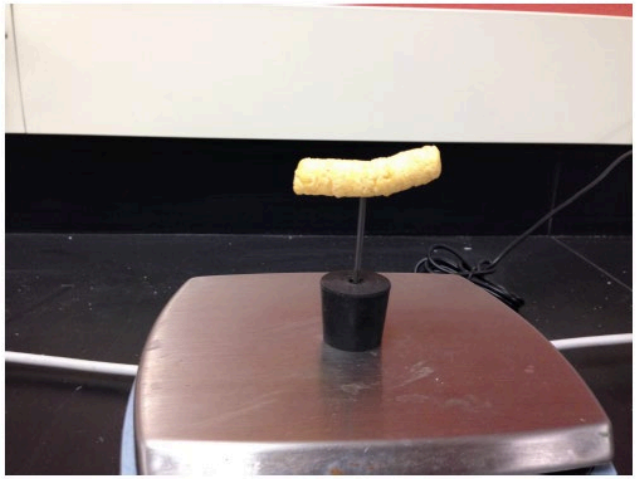
In this lab you will use the specific heat of water and its change in temperature to determine the caloric content in a Frito™ and Cheeto™.



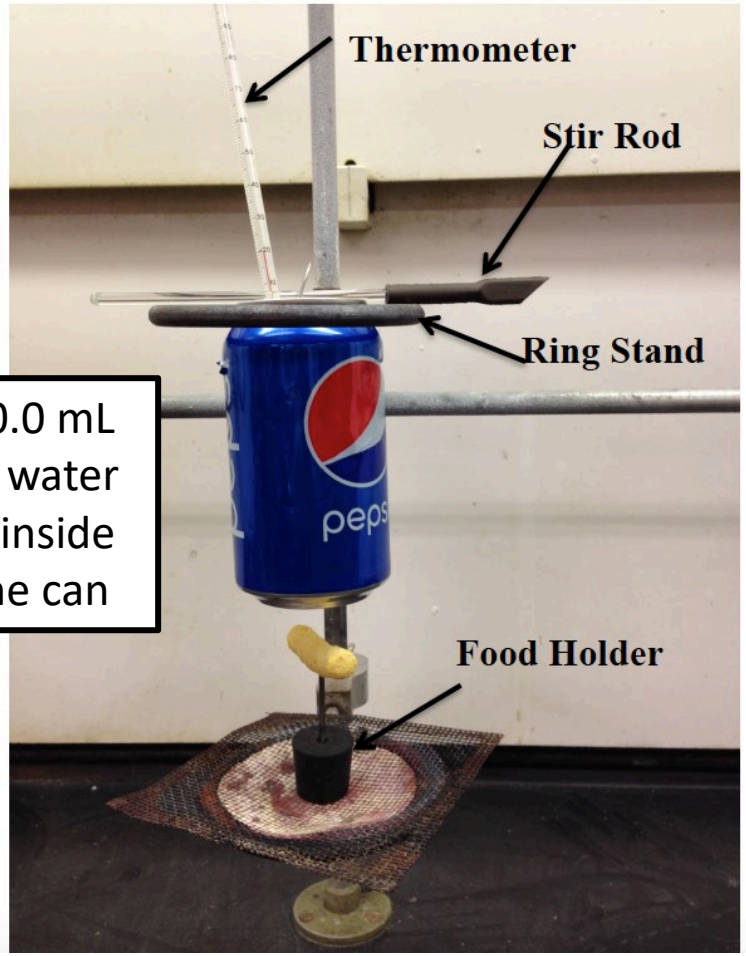
- This part of the lab will be done in the hood.
- You will do this twice, once with Fritos and once with Cheetos.

Part B setup – Food is fuel

1. Measure the mass of one of the food samples



2. Setup the calorimeter

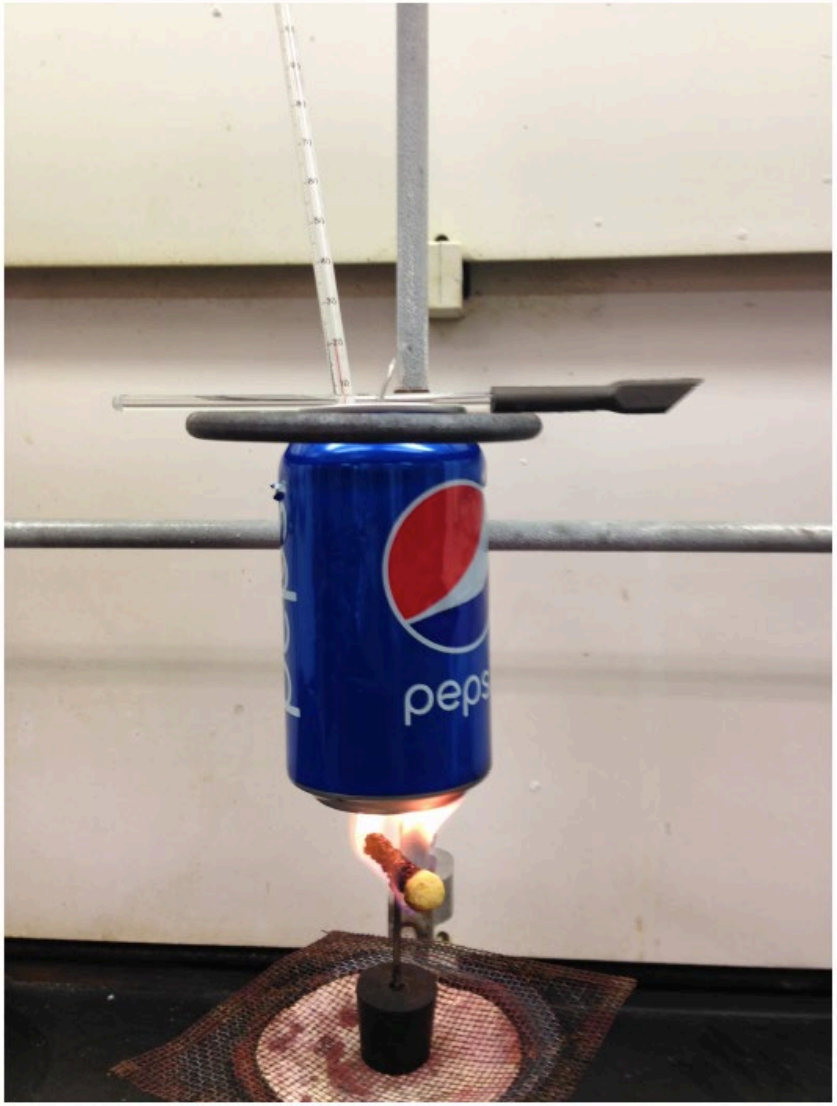


3. Record the initial temperature of the solution. This is **$T_{initial}$ for the solution of water.**

Part B setup – Food is fuel

4. Light the food sample on fire with the Bunsen burner. Allow all of the food to burn and then record the maximum temperature that the water reaches. This is the **T_{final} for the solution of water.**

5. Determine the mass of remaining food sample.



Part B setup – Calculations

3. You know have everything you need to calculate the heat (q) released by the food sample.

$$q_{water} = m_{water} \, x \, C_{s(water)} \, x \, \Delta T_{(water)}$$

$$q_{water} = -q_{food}$$

4. Use the q_{food} to calculate the energy content per gram of the food sample (see the example calculations in manual)

What is due and when?

- An electronic notebook upload of your completed Calorimetry/Thermochemistry Lab is due through the Canvas Lab page by this Wednesday at 11:59pm
  - CamScanner is a great free app that you can use
    - The Basic package is free and allows you to take pictures of multiple pages and convert to a single pdf file for upload
  - This upload must be a pdf!
- Be sure to complete your Intermolecular Forces Pre-lab assignment BEFORE lab next week!